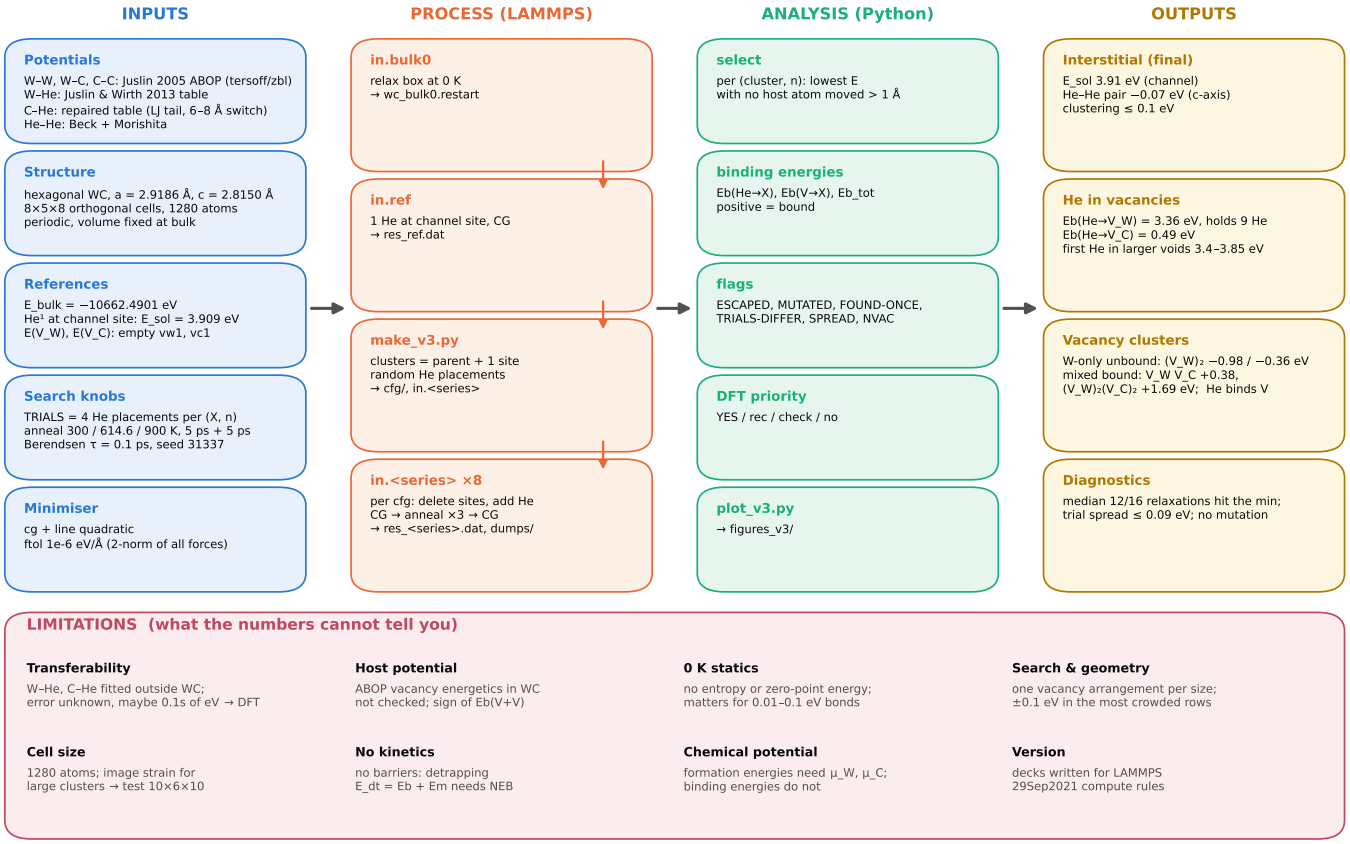
He in WC: quick-reference sheet

Model schema, LAMMPS commands and their knobs, every equation in the analysis scripts, the order of events in every input file, and the terms. Values are those of the v3 binding-energy package (`he_be_v3`), the formation-energy package (`wc_ef`) and the Juslin & Wirth 2013 (JW) benchmark. Sign convention throughout: **positive binding = bound**.

1 Model schema (inputs, results, limitations) · 2 Workflow and files · 3 Order of events in every input file · 4 LAMMPS commands and knobs · 5 Equations · 6 Glossary · 7 Renders of the simulation cells. Schematics (`schematics/S1--S6`), renders (`renders/R1--R4`) and animations (`gifs/G1--G4`) are also separate files.

He in WC: model schema

every reported energy is a 0 K conjugate-gradient minimum; MD only moves the system between minima



1 Model schema

Full machine-readable version: `model_schema.yaml`. Sensitivity = how much the reported binding energies move if the input is changed: **high** > 0.1 eV or changes a conclusion; **med** 0.01–0.1 eV; **low** < 0.01 eV.

Input	Value used	Source / note	Sens.
W–W, W–C, C–C	ABOP, `tersoff/zbl`, file `WCH.tersoff`	Juslin *et al.* J. Appl. Phys. 98 (2005) 123520. He is `NULL` in it (closed shell).	high
W–He	pair table `WHe.table`, keyword `WHe`, 1e-6–3.25 Å	Juslin & Wirth, J. Nucl. Mater. 432 (2013) 61; same data as LAMMPS `W_He_JW2013.table`. Validated in bcc W only.	high
C–He	`CHe.table`, 0.005–8 Å	group table; LJ tail (σ 2.524 Å, ε 16.0 meV) beyond 3.6 Å, switched to 0 at 6–8 Å; core extended < 0.1 Å. Not validated.	high
He–He	`HeHe.table`, 0.00076–3.8 Å	Beck (1968) + Morishita (2003) core. fcc He -0.00538 eV/atom (JW: 0.0054).	med
Host lattice	hexagonal WC, $a = 2.91862$, $c = 2.81498$ Å ($c/a = 0.9645$), $E_{\mathrm{coh}} = -8.33007$ eV/atom	the potential's own 0 K values from `in.bulk0`	none
Cell	$8 \times 5 \times 8$ orthogonal cells, 1280 atoms, $23.35 \times 25.28 \times 22.52$ Å, `p p p`	fixed volume for all defect cells (constant-volume convention)	med
E_{bulk}	-10662.4900860 eV	checkpoint; must agree to $\sim 10^{-6}$	–
He reference	one He at the trigonal-prismatic channel, $(\frac{1}{2}, \frac{5}{6}, \frac{1}{2})$ of cell (4,2,4); $E_{\mathrm{sol}} = 3.909143$ eV	checkpoint from `in.ref`	–
Minimiser	`cg`, `line quadratic`, etol 0, ftol 10^{-6} eV/Å, 100000 / 1000000	ftol is the 2-norm of the whole force vector	low
Search	TRIALS 4; anneal 300 / 614.6 / 900 K; 5000 + 5000 steps of 1 fs; Berendsen τ 0.1 ps; seed 31337	4 relaxations per trial; lowest intact minimum kept	high
He placement	within 1.4 Å (W site) / 0.9 Å (C site) of a vacated site; He–He and He–host ≥ 1.5 Å	constraints relaxed stepwise only when the void is crowded	low
Clusters	`vw1 vw2 vw2a vw3 vc1 vwc vwc2 vw2c2`; He 0–12 (vc1 0–4, vwc/vwc2 0–10)	each = parent + one site (Sec. 5.3)	–
Diagnostics	host atom moved > 1 Å = mutated; core = farthest site +2.2 Å; same min < 1 meV; spread warn 0.10 eV; R_g jump 1.5 Å	`analyze_v3.py`	–

Results: binding energies (v3, Oct 2026) – what to expect on a rerun

Quantity	Value	Comment
E_{sol}, channel site	3.909 eV	lowest interstitial site in WC
E_{sol}, metastable site near C	7.302 eV	
He–He pair $E_{\mathrm{int}}(d)$	-0.070 (2.815 Å, same channel), -0.036 (2.92 Å), $+0.018$ (4.05 Å), $\lvert E\rvert < 0.003$ beyond 5 Å	negative = attractive
$E_{\mathrm{b}}(\mathrm{He} \to \mathrm{He}_n)$ interstitial	0.07 (chain), 0.02–0.10 (bundle)	weak; most sensitive to 0 K / entropy
$E_{\mathrm{b}}(\mathrm{He} \to V_{\mathrm{W}})$	3.36 eV	DFT check needed
$E_{\mathrm{b}}(\mathrm{He} \to V_{\mathrm{C}})$	0.49 eV; further He 0.05–0.35 eV	extra He sit in the surrounding channels
$E_{\mathrm{b}}(\mathrm{He} \to \mathrm{He}_{n-1}V_{\mathrm{W}})$	1.82, 1.57, 1.62, 1.23, 1.28 (n=2–6); 0.12 (n=7); 0.42, 0.53; -0.18, -0.14 (n=10, 11)	capacity 9 He; the n=7 dip is reproducible
First He into larger clusters	3.85 (c-axis V_2, V_3), 3.41 (in-plane V_2), 3.41–3.57 (mixed)	all below $E_{\mathrm{sol}} = 3.91$
He filling of multi-vacancy clusters	stays positive in the range studied	min 0.91 (c-axis V_2, 5.5 He/V); 0.29 ($V_{\mathrm{W}}V_{\mathrm{C}}$, 4 He/V)
Empty W clusters	-0.98 (c-axis V_2), -0.36 (in-plane V_2), -1.29 (V_3)	unbound
Empty mixed clusters	$+0.38$ ($V_{\mathrm{W}}V_{\mathrm{C}}$), $+0.42$ ($V_{\mathrm{W}}(V_{\mathrm{C}})_2$), $+1.69$ ($(V_{\mathrm{W}})_2(V_{\mathrm{C}})_2$)	bound; DFT check of $V_{\mathrm{W}}V_{\mathrm{C}}$ first
Preferred W divacancy	in-plane for $n \leq 2$ (by 0.62, 0.18, 0.11); c-axis from $n = 3$ (by 0.13–1.07)	He switches the orientation
$E_{\mathrm{b}}(V \to \mathrm{He}_n V_m)$	W clusters: > 0 from $\sim$1 He/V, 12.4 at 6 He/V; V_{W} to clusters with V_{C}: 3.3–9.2; V_{C}: 0.04–2.1	He stabilises vacancy clusters
Trap mutation	none; max host displacement 0.39 Å	He escapes to channels instead
Search	median 12/16 hits, trial spread $\leq$0.09 eV; v2 reproduced within 0.04 eV	crowded rows $\pm$0.1 eV

Results: formation energies (`wc_ef`, Oct 2026)

Reference phases (same potential; 0 K, zero stress). They set the window $\mu_C \in [\varepsilon_{WC} - \varepsilon_W,\ \varepsilon_C] = [-7.7700,\ -7.3612]$ eV, width 0.409 eV.

Phase	Energy	Relaxed lattice	Meaning
hexagonal WC	$\varepsilon_{WC} = -16.660141$ eV/f.u.	a 2.91862, c 2.81498 Å	equals $E_{bulk}/640$ (check passed)
bcc W	$\varepsilon_W = -8.890096$ eV/atom	a 3.16496 Å	W-rich limit
graphite (= graphene)	$\varepsilon_C = -7.361187$ eV/atom	a 2.5162 Å (C–C 1.453 Å)	C-rich limit (no interlayer term)
diamond	−7.316044 eV/atom	a 3.56012 Å	45 meV/atom above graphite
W2C (CdI2 type)	−23.958573 eV/f.u.	a 2.9853, c 5.0407 Å	**+1.59 eV above WC + W**; $\Delta H_f = +1.18$: unstable in this potential
rocksalt WC	−15.676602 eV/f.u.	a 4.38317 Å	+0.98 eV/f.u. above hexagonal WC
ΔH_f(WC)	−0.4089 eV/f.u.		expt. ≈ −0.42

Point defects ($\Delta E = E(\text{defect}) - E_{bulk}$; E_f in eV). "Elemental refs" uses bcc W and graphite at the same time, the convention of Kong *et al.* (J. Nucl. Mater. 2010, GGA, 128 atoms), so those two columns compare like with like.

Defect (lowest candidate)	ΔE	W-rich	C-rich	elemental refs	DFT	Note
V_W	12.5604	3.670	3.261	3.670	4.14	same as v3
V_C	8.2931	0.523	0.932	0.932	0.39	cheapest defect everywhere in the window
C_i (channel site)	−4.5468	3.223	2.814	2.814	3.41	the y-dumbbell relaxes to the same state; a-dumbbell +0.79; W-layer sites +1.73 and +5.76
W_i (dumbbell along a, relaxed)	3.4459	12.336	12.745	12.336	11.58	still dropping at cycle 4; irrelevant thermally
W_C (W on C site)	8.0954	9.215	10.033	9.624	8.69	stays on site
C_W (C on W site)	7.8708	6.751	5.933	6.342	8.81 (5.29 as $C_i + V_W$)	relaxes toward the channel: a V_W–C_i pair bound by 0.14 eV
V_C pair, c-axis	17.0469	1.507	2.325			binding −0.46: C vacancies **repel**
V_C pair, in-plane	16.8381	1.298	2.116			binding −0.25: repel
$V_W V_C$	20.4731	3.813	3.813			binding +0.38 (same as v3)

μ-independent combination	This work	DFT (Kong)	Comment
Schottky pair $V_W + V_C$ (isolated)	4.193	$4.53 + \Delta H_f \approx 4.1$	agrees; the split between W and C differs by $\sim$0.5 eV each way
C Frenkel pair $V_C + C_i$	3.746	3.80	agrees
W Frenkel pair $V_W + W_i$	16.006	15.72	agrees within 0.3
antisite pair $W_C + C_W$	15.966	17.50	DFT's C_W (8.81) relaxes to $C_i + V_W$ at 5.29 (sum 13.98)

Conditions (equilibrium site fractions, no entropy, $g = 1$ except pairs):

Condition	$\mu_C - \varepsilon_C$	V_W	V_C	C_i	C/W at 1000 K / note
W-rich limit (WC/W)	-0.409	3.670	0.523	3.223	0.99769 (0.23% C-poor)
mid-window	-0.204	3.466	0.728	3.019	0.99978
C-rich limit (WC/graphite)	0	3.261	0.932	2.814	0.99998
50/50 WC, 300–1500 K	0 (clamped)	3.261	0.932	2.814	exact 50/50 is out of reach: V_C always outnumbers C_i; closest = C-rich limit (C/W 0.99926 at 1500 K)
$WC_{0.999}$, 1000 K	-0.337	3.598	0.595	3.151	inside the window
$WC_{0.99}$, 1000 K	-0.536	3.797	0.396	3.350	0.127 eV beyond the W-rich limit: non-equilibrium
$WC_{0.95}$, 1000 K	-0.677	3.938	0.255	3.491	0.268 eV beyond: non-equilibrium

He_nV_m formation energies at the same conditions: `he_v_formation.csv` ($E_f = E - E_{bulk} + n_W\mu_W + n_C\mu_C$, He from vacuum), e.g. He_1V_W 4.22 (W-rich) / 3.81 (C-rich); clusters with equal W and C vacancies (V_WV_C, $(V_W)_2(V_C)_2$) do not depend on μ.

Take-aways. (1) ΔH_f(WC) and all μ-independent combinations agree with experiment/DFT. (2) The potential puts W2C far above WC + W, so the W-rich limit is bcc W, and a C-depleted layer would separate into WC + W rather than form W2C. (3) V_C is the dominant defect (0.5–0.9 eV), so WC is always slightly C-poor; C vacancies repel each other, so they stay dispersed. (4) C_i sits in the same channel site as He.

Limitations

- **Transferability.** W–He and C–He pair potentials were not fitted in WC. Error unknown, possibly tenths of an eV. → DFT on ∼12–15 small cells (≤ 2 vacancies, ≤ 4 He).
- **Host potential.** ABOP vacancy and divacancy energetics in WC unchecked; controls the sign of $E_b(V{+}V)$. → compare V_W, V_C, V_WV_C with DFT.
- **0 K statics.** No vibrational entropy, no zero-point energy. Matters for 0.01–0.1 eV bonds (He–He), barely for eV-scale trapping.
- **Search.** Stochastic; can miss the global minimum in crowded clusters. Shows as `TRIALS-DIFFER` / `FOUND-ONCE`; v3: ±0.1 eV in the most crowded rows.
- **Cluster geometry.** One vacancy arrangement per cluster size; only the He were searched. Other arrangements (mixed clusters) may bind more.
- **Cell size.** One 1280-atom cell; image strain grows with cluster size. → repeat key rows at $10 \times 6 \times 10$.
- **Chemical potential.** Vacancy *formation* energies need μ_W, μ_C; every *binding* energy here is independent of them.
- **No kinetics.** No barriers, so no detrapping energy $E_{dt} = E_b + E_m$ or desorption temperature. → NEB.
- **Version.** Decks follow the 29 Sep 2021 compute-currency rules; newer LAMMPS is more permissive, so they still run.
- **Formation energies.** One W2C structure; graphite = graphene (no interlayer term); dilute defects, no formation entropy ($g = 1$); equilibrium only – a sputter-depleted surface is outside it.

2 Workflow and files

Step	What it does	Writes
`in.bulk0`	builds the $8 \times 5 \times 8$ WC cell and relaxes the box at 0 K	`wc_bulk0.restart`; prints E_{bulk}, a, c
`in.ref`	one He at the channel site, CG	`res_ref.dat` (E_{bulk}, $E(\mathrm{He}_1)$)
`make_v3.py`	builds clusters (parent + one site), random He placements (TRIALS per n), one deck per series	`cfg/<series>_<k>.lmp`, `cfg/index.txt`, `in.<series>`, `series.json`
`in.<series>`	per cfg: delete sites, add He, CG; anneal 300/614.6/900 K, quench, CG after each	`res_<series>.dat` (`k n m trial nvac cycle E dmax nfar kin rg rg0`), `dumps/`
`analyze_v3.py`	lowest intact minimum per (cluster, n), binding energies, flags, DFT column	`binding_v3.csv`, `clusters_v3.csv`
`plot_v3.py`	figures	`figures_v3/`
`run.slurm`, `submit_v3.sh`	12 MPI tasks, `short` partition, bulk0 first, the rest `afterok`	`log.<deck>`
`check_he_positions.py`	where each He sits in the reported minima (void / channel / core)	printed table
Formation energies (`wc_ef`)		
`make_ef.py`	reference crystals, 19 defect cells, decks (`--cell` for a size check)	`phases/`, `cfg/def_<k>.lmp`, `in.*`, `defects.json`
`in.bulk_ef`	bulk cell (identical to `in.bulk0` for $8 \times 5 \times 8$)	`wc_bulk_ef.restart`, `res_bulk.dat`
`in.phases`	bcc W, graphite, diamond, W2C, hex WC, rocksalt WC, box-relaxed	`res_phases.dat`
`in.defects`	19 defect cells, CG + 3-step anneal ladder	`res_defects.dat`, `dumps/`
`analyze_ef.py`	μ window, formation energies by condition, 50/50 and WC_{1-y} solves	`ef_summary.txt`, `ef_conditions.csv`, `ef_candidates.csv`, `ef_vs_mu.png`, `he_v_formation.csv`
`submit_ef.sh`	bulk_ef and phases at once; defects `afterok` bulk_ef	`log.*`
Renders (`wc_he_reference/scripts`)		
`renders.py`	R1–R4 below (OVITO, Tachyon)	`renders/*.png`
`render_dump.py`	any dump from your runs, same style, vacancies from its cfg	one png

Run analysis only when `squeue -u $USER` is empty and `grep -l "^ERROR" log.*` prints nothing.

3 Order of events in every input file

What each script and deck does, step by step, in the order it happens. Decks are LAMMPS input files (run with `lmp -in`); `.py` files run on a login node or your laptop. Files a step reads are blue; files it writes are orange.

3.1 Binding energies (he_be_v3)

Run order: `make_v3.py` (already run) → `submit_v3.sh`: `in.bulk0` → (`in.ref` and the eight `in.<series>` in parallel) → `analyze_v3.py` → `plot_v3.py`, `check_he_positions.py`.

`make_v3.py`

1. List the perfect-lattice W, C and channel sites of the $8 \times 5 \times 8$ cell in cell units.
2. Build each cluster: seed site W0 = (4.5, 2.5, 4) or C0 = (4.5, 3.167, 4.5); add one site at a time (compact rule; `vw2a` restricted to the basal plane).
3. For each series and n: TRIALS random He placements inside the void (one for $n = 0$); centre and radius of the core sphere.
4. Write each cfg: `variable n m trial`; a `region sphere` per vacated site, `group vac`, `nvac`, `delete_atoms`; `region core`; one `create_atoms 3 single` per He → `cfg/<series>_<k>.lmp`, `cfg/index.txt`.
5. Fill the deck template (setup, record, 3× anneal + record, loop tail) → `in.<series>`, `in.ref`, `series.json`.

`in.bulk0`

1. `units metal`, `atom_style atomic`, `boundary p p p`.
2. `lattice custom` (orthogonal WC basis, 614.6 K guess for a, c); `region`/`create_box 3`/`create_atoms` (basis 1–2 W, 3–4 C); masses.
3. `pair_style hybrid/overlay` + four `pair_coeff`; neighbour list.
4. `fix box/relax ... couple xy` + CG; `unfix`; CG again.
5. Print a, c, E_{bulk} → `wc_bulk0.restart`.

`in.ref`

1. `wc_bulk0.restart`; pair lines; `run 0` → E_{bulk}.
2. `create_atoms 3 single` at the channel site (4.5, 2.833, 4.5); CG; `run 0`.
3. Print E_{bulk}, $E(\mathrm{He}_1)$, E_{sol} → `res_ref.dat`, `dump.ref.lammps`.

`in.<series>` (e.g. `in.vw1`)

1. Once: `wc_bulk0.restart`, pair lines, `run 0` → E_{bulk}; header lines of `res_<series>.dat`; `mkdir dumps`.
2. Loop over k: `clear`; `wc_bulk0.restart`; pair, neighbour, `timestep 0.001`, `cg` + `line quadratic`.
3. `include cfg/<series>_<k>.lmp`: delete the vacated atoms, define core, add the He.
4. Groups `host`, `heg`; R_{g0}; computes `dsp`, `dmax`, `nfar`; `thermo_style` with them.
5. Cycle 1: CG → `run 0` → print `k n m trial nvac 1 E dmax nfar kin rg rg0` → dump.
6. Cycles 2–4 (T = 300, 614.6, 900 K): `velocity create` (`loop geom`) → `nve` + Berendsen at T, 5000 steps → Berendsen $T \to 1$ K, 5000 steps → unfix → CG → `run 0` → print → `dumps/<series>_<k>_c<cycle>.lammps`.
7. `next k`; `jump SELF cfg`.

`analyze_v3.py`

1. Read `res_ref.dat`, `series.json`, every `res_<series>.dat`.
2. Per (series, n): lowest intact minimum; hits, trial spread, winning cycle; flags.
3. $E_b(\mathrm{He} \to)$, $E_b(V_s \to)$ for each parent, E_b^{tot}; DFT column.
4. Print tables (and v3 − v2 with `--v2`) → `binding_v3.csv`, `clusters_v3.csv`.

`plot_v3.py`: `binding_v3.csv` → `figures_v3/`. `check_he_positions.py`: same minimum as the analysis → its dump → each He classified as void / channel / other, in core or not.

3.2 Formation energies (wc_ef)

Run order: `make_ef.py` (already run) → `submit_ef.sh`: `in.bulk_ef` and `in.phases` at once, `in.defects` after `in.bulk_ef` → `analyze_ef.py --v3 ../he_be_v3/binding_v3.csv`.

`make_ef.py`

1. Write `in.bulk_ef` for the chosen cell (default $8 \times 5 \times 8$).
2. Build six reference crystals; write `phases/<name>.data` and an include `phases/ph_<k>.lmp` with its relaxation style (`iso`; hexagonal `x y z couple xy`; graphite `x y couple xy`, z fixed) → `in.phases`.
3. List the 19 defects as removed sites + added atoms, around cell (4, 2, 4).
4. Write each cfg: `dname`; one site → `group vac region v1`, two → `region union`; `delete_atoms`; `group host`; `create_atoms`; `group added subtract all host` → `cfg/def_<k>.lmp`.
5. Fill the deck template (v3 ladder) → `in.defects`, `defects.json`.

`in.bulk_ef`: as `in.bulk0` → `wc_bulk_ef.restart`, `res_bulk.dat`.

`in.phases`

1. Header of `res_phases.dat`; loop over k.
2. `clear`; `include phases/ph_<k>.lmp`: `read_data phases/<name>.data`, pair lines, groups `w`, `c`.
3. `fix box/relax` + CG, twice; plain CG; `run 0`.
4. Print `name nW nC E lx ly lz press`; dump.

`in.defects`

1. Once: `wc_bulk_ef.restart`; E_{bulk} → header of `res_defects.dat`.
2. Loop: `clear`, restart, pair, `include cfg/def_<k>.lmp`.
3. Computes `dsp`/`dmax`/`nfar` on the original host atoms.
4. Cycles 1–4 exactly as `in.<series>`; each prints `k name cycle nvac E dmax nfar` and `xcm(added)` → `dumps/def_<k>_c<cycle>.lammps`.

`analyze_ef.py`

1. `res_phases.dat`: ε_{W}, ε_{C} (lower of graphite, diamond), $\varepsilon_{\text{W}_2\text{C}}$; checks hex WC and rocksalt; $\varepsilon_{\text{WC}} = E_{\text{bulk}}/640$; ΔH_f.
2. Window: $\mu_{\text{C}}^{\max} = \varepsilon_{\text{C}}$; $\mu_{\text{W}}^{\max} = \min(\varepsilon_{\text{W}}, \varepsilon_{\text{W}_2\text{C}} - \varepsilon_{\text{WC}})$ if W2C is stable.
3. `res_defects.dat`: lowest minimum per candidate (vacancies: intact only); C_i, W_i = lowest candidate.
4. μ-independent pairs and pair bindings.
5. Conditions: W-rich, C-rich, mid; 50/50 at each T (bisection on C/W = 1, clamped to the window); WC_{1-y} at `--Tdef`.
6. `ef_summary.txt`, `ef_conditions.csv`, `ef_candidates.csv`, `ef_vs_mu.png`; with `--v3`: `he_v_formation.csv`.

4 LAMMPS commands: what they do here and how to turn the knobs

↑/↓ = increasing/decreasing the knob. "Cost" = wall time.

Command (as used)	Role in this model	Knob → impact
Setup		
`units metal`	eV, Å, ps, K, bar, amu	none; every table and tolerance is in these units
`atom_style atomic`	positions, types, no charges	–
`boundary p p p` `change_box all boundary p p p`	fully periodic bulk (re-asserted after `read_restart`)	`p p f` would add a surface: *not* a bulk binding energy
`lattice custom a a1 1 0 0 a2 0 ${blat} 0 a3 0 0 ${clat} basis ...`	orthogonal WC cell: W at $(0,0,0)$, $(\frac{1}{2},\frac{1}{2},0)$; C at $(0,\frac{2}{3},\frac{1}{2})$, $(\frac{1}{2},\frac{1}{6},\frac{1}{2})$	a, c are only a starting guess (box/relax finds the 0 K values); wrong basis = wrong phase
`lattice bcc 3.1652` (benchmark)	bcc W, Ackland–Thetford a_0	must be the potential's own a_0 or the cell is strained
`read_data phases/<name>.data` (`wc_ef`)	reads a reference crystal written by `make_ef.py` (3 atom types, `Atoms # atomic`)	starting lattice only; box/relax finds the potential's own
`region box block 0 8 0 5 0 8` `create_box 3 box, create_atoms`	$8 \times 5 \times 8$ cells, 3 types (W, C, He)	bigger ↑: less image interaction, cost $\propto N$; keep every edge $> 2\times$ the longest cutoff; first thing to test for large clusters
`mass 1 183.85` etc.	masses	affect MD speed only, never a 0 K energy
Interactions		
`pair_style hybrid/overlay` `tersoff/zbl table linear 10000`	ABOP for the host, tables for every He pair, summed	`linear` → `spline`: smoother forces; table points N ↑: less interpolation noise (CG reaches ftol more easily), more memory; N ↓ below ∼1000: force noise, CG stalls
`pair_coeff * * tersoff/zbl` `WCH.tersoff W C NULL`	maps types 1,2 to W, C; He excluded from bond order	order must match the type numbering; dropping `NULL` puts He in the ABOP (wrong)
`pair_coeff 1 3 table WHe.table WHe` (and `2 3`, `3 3`)	W–He, C–He, He–He	optional final argument = cutoff; default is the table's last point

Command (as used)	Role in this model	Knob → impact
`neighbor 2.0 bin`	skin added to the cutoff	skin ↑: fewer rebuilds, more pairs per step; skin ↓: risk of missed pairs during hot MD unless `check yes`
`neigh_modify every 1 delay 0 check yes`	rebuild whenever an atom moved > skin/2	safest setting; `delay` ↑ is faster but can miss pairs at 900 K
Minimisation (every reported energy)		
`min_style cg`	Polak–Ribière conjugate gradient	`fire`: more robust from bad starts (overlapping He), usually slower to tight ftol; `sd`: slow; `hftn`: memory heavy
`min_modify line quadratic`	line search that uses forces near the minimum	reaches ftol 10^{-6} where the default `backtrack` stalls on round-off; `forcezero` is the alternative
`minimize 0.0 1.0e-6 100000 1000000`	etol, ftol, maxiter, maxeval	**etol** = relative energy change; 0 disables it so ftol decides. **ftol** = $\|\mathbf{F}\|_2$ of all $3N$ components (eV/Å), *not* the max force: ↓ tighter, more iterations, energy changes by $\ll$1 meV below 10^{-4}. **maxiter/maxeval**: if hit, the energy is not converged (check "Stopping criterion" in the log)
`fix rel all box/relax x 0 y 0 z 0 couple xy vmax 0.001` (`in.bulk0` only)	relaxes cell edges to zero stress; `couple xy` keeps one hexagonal a	`vmax` ↑: faster, can oscillate; `couple none` lets $L_x \neq L_y$ (breaks $\sqrt{3}$ ratio); re-run `minimize` after `unfix` because the box energy term is gone
`fix rel all box/relax iso 0.0 vmax 0.001` (cubic phases) `fix rel all box/relax x 0.0 y 0.0 couple xy vmax 0.001` (graphite)	zero-stress reference crystals; graphite keeps z fixed because the ABOP has no interlayer term	`z 0.0` on graphite would let L_z drift (no force across layers)
`fix hold grp setforce 0.0 0.0 0.0` (benchmark)	holds an atom (octahedral SIA, He at the tet–tet midpoint for E_m)	removes those atoms from the force norm; a constrained, not a true, minimum
Search MD (never an energy)		
`timestep 0.001`	1 fs	↑ 2 fs: faster, energy drift with light He at 900 K; ↓: slower, no benefit for a search
`velocity all create T seed rot no mom yes dist gaussian loop geom`	initial velocities at T	**seed**: different trajectory (independent trials); **loop geom**: velocities from positions, so identical for any MPI count and for non-contiguous IDs (needed after `delete_atoms`); `mom yes` zero net momentum
`fix nve all nve` + `fix tb all temp/berendsen T T 0.1`	anneal at constant T	**T** ↑: crosses larger barriers, more He escape / host damage; ↓: stays near the start. **Tdamp** ↓ (<0.05 ps): stiff, unphysical coupling; ↑ (>1 ps): slow to reach T
`fix tb all temp/berendsen T 1.0 0.1`	quench: ramps target $T \to 1$ K	longer quench ↑: settles into a deeper basin before CG
`run 5000`	5 ps anneal / quench	↑: more barrier crossings per trial, cost linear; $3 \times (5 + 5)$ ps per trial
`run 0 post no`	refreshes thermo/computes without moving atoms	required before reading a compute in a variable between runs (29 Sep 2021 rule)
Building each configuration		

Command (as used)	Role in this model	Knob → impact
`region v1 sphere x y z 0.5 units box` `region vall union 2 v1 v2`	selects the atoms to remove	radius must be < nearest-neighbour distance (W–C 2.196 Å) or two atoms go
`group vac region vall`; `$(count(vac))`	atoms that will be deleted	`nvac` must equal the number of sites (flag `NVAC`)
`delete_atoms group vac compress no`	makes the vacancies; keeps IDs	`compress yes` renumbers: breaks ID-based tracking
`create_atoms 3 single x y z units box`	inserts one He	–
`create_atoms 1|2 single ...` (`wc_ef`)	inserts the interstitial / antisite / dumbbell atoms	dumbbell half-length: C 0.70, W 1.10 Å
`group host type 1 2` *before* `create_atoms`; `group added subtract all host`	static groups: the new atoms are not in `host`	displacement statistics stay on the original lattice atoms
`region vall union N ...`	union of several vacancy spheres	needs $N \geq 2$: a single site uses `group vac region v1` directly
`region core sphere x y z r units box`	"cluster" region for `count(heg,core)`	r = farthest site +2.2 Å; ↑: escaped He still counted; ↓: He at the void edge miscounted
`read_restart wc_bulk0.restart`	starts every cfg from the same relaxed bulk	pair coefficients of `hybrid/table` are re-issued after it
`write_restart` (benchmark `in.vgen`)	saves V_m for the next step	re-initialises: computes become stale, so read them into variables *before* it
`clear`	resets the whole system for the next cfg	everything after it must be redefined
Measurements		
`compute dsp host displace/atom`	displacement of each W/C atom from where it was when the compute was defined (perfect lattice)	–
`compute dmax host reduce max c_dsp[4]`	largest host displacement (`dmax`)	–
`variable farv atom (c_dsp[4]>1.0) + reduce sum`	host atoms moved > 1 Å (`nfar`)	threshold ↓ (0.7 Å): flags strongly relaxed but intact neighbours; ↑ (1.5 Å): misses Frenkel-pair mutations
`compute pe/atom`, `reduce max` (benchmark)	finds the highest-energy W atom (JW vacancy rule)	ties broken by lowest ID
`gyration(heg)`, `count(heg,core)`, `xcm()`	He radius of gyration, He inside core, centre of mass	–
`$(xcm(added,x))` (`wc_ef`)	where the added atom(s) ended up	0 for vacancy cells (empty group)
`$(count(w))`, `$(count(c))` (`in.phases`)	atoms of each species in a phase	energies per atom / per f.u.

Command (as used)	Role in this model	Knob → impact
`thermo_style custom step temp pe press c_dmax c_nfar`	makes the computes current at every output	a compute read between runs *must* appear here
`thermo_modify norm no`	totals, not per-atom energies	`norm yes` would divide every energy by N
Script control		
`variable k loop 49 pad` / `label cfg` / `next k` / `jump SELF cfg`	flat loop over cfg files	`jump SELF` needs `lmp -in deck` (not `< deck`); `pad` gives 01, 02, . . .
`variable k index 01 49`	test only cfgs 01 and 49	quick syntax test
`variable dname string Ci_ch`	names the defect in the result line	–
`print "..." file f` vs `append f`	`file` starts the file, `append` adds a line	a second `file` would erase earlier results
`$(expr)`, `${v}`, `v_name`, `$(pe:%.10f)`	immediate evaluation, substitution, reference, formatted output	`$()` is evaluated when the line is read
`include cfg/vw1_${k}.lmp`	pulls in one configuration	–
`print "..." append res_vw1.dat`	one result line per relaxation	–
`write_dump all custom f id type x y z modify sort id`	snapshot of each minimum	for OVITO; not used by the analysis
`shell mkdir dumps`	output folder	–

Script knobs

Knob (file)	Value	Turn it → effect
`TRIALS` (make_v3)	4	↑: fewer missed minima (`TRIALS-DIFFER`, `FOUND-ONCE` disappear), cost linear; 3 if the 3 h limit is tight
`TEMPS` (make_v3)	300, 614.6, 900	add 1200 K: crosses bigger barriers but more He escape / host damage; drop 900 K: misses deep rearrangements
`N_HEAT`, `N_QUENCH`	5000, 5000	↑: more sampling per trial; 3000 + 3000 cuts the cost 40%
`TAU`	0.1 ps	Berendsen τ; see `temp/berendsen` above
`SEED`	31337	changes every placement and velocity; rerun with a new seed to test convergence
`NX, NY, NZ`	8, 5, 8	↑: smaller image interaction; regenerate `in.bulk0` too
`R_SITE`	W 1.4, C 0.9 Å	↑: He placed off-centre, more crowding with the host
`HE_HE_MIN`, `HE_HOST_MIN`	1.5, 1.5 Å	↓: denser starts, risk of huge initial forces (CG can eject atoms)
core radius	farthest site + 2.2 Å	sets what counts as "escaped"
mutation threshold (deck)	1.0 Å	see `farv` above
`TOL` (analyze_v3)	10^{-3} eV	what counts as reaching the same minimum (`hits`)
`SPREAD_WARN`	0.10 eV	spread of trial minima above which `TRIALS-DIFFER` is printed

Knob (file)	Value	Turn it → effect
`RG_JUMP`	1.5 Å	$R_g - R_{g0}$ above which `SPREAD` is printed
Formation energies		
`--cell` (make_ef)	8 5 8	10 6 10 for the size check (new folder, new bulk run)
`TEMPS`, `N_HEAT`, `N_QUENCH`, `SEED` (make_ef)	as v3, 4242	the anneal ladder for the defect cells
`--T` (analyze_ef)	300 600 1000 1500	temperatures of the 50/50 solve
`--deficit`, `--Tdef`	0.001 0.01 0.05; 1000 K	C deficits y for WC_{1-y}
`G` (analyze_ef)	1 (pairs 3, 6)	site multiplicity: concentrations $\times g$, E_f unchanged

29 Sep 2021 rules the decks follow

- A compute used in a variable between runs must be current: put it in `thermo_style` and read it straight after a `run 0`.
- `write_restart` re-initialises the system: read every compute into a variable before it.
- `jump SELF` only works with `-in`; one file per case, flat loop, no `if/else`, no `&` continuations.
- After `delete_atoms ... compress no` IDs have gaps, so `velocity ... loop geom`.

5 Equations

$E(X)$ = relaxed total energy of a cell containing defect X (same box as $E_{\rm bulk}$). He_1 = one interstitial He at the reference site. X = vacancy cluster with $n_{\rm W}$ W and $n_{\rm C}$ C sites; X' = its parent (one site fewer).

5.1 WC (`analyze_v3.py`)

$$E_{\rm sol} = E(\mathrm{He}_1) - E_{\rm bulk} \tag{1}$$

$$E_{\rm b}(\mathrm{He} \to \mathrm{He}_n X) = E(\mathrm{He}_{n-1}X) + E(\mathrm{He}_1) - E(\mathrm{He}_n X) - E_{\rm bulk} \tag{2}$$

$$E_{\rm b}(V_s \to \mathrm{He}_n X) = E(\mathrm{He}_n X') + E(V_s) - E(\mathrm{He}_n X) - E_{\rm bulk}, \qquad E(V_s) = E(\text{empty vw1 or vc1}) \tag{3}$$

$$E_{\rm b}^{\rm tot}(\mathrm{He}_n X) = n_{\rm W}E(V_{\rm W}) + n_{\rm C}E(V_{\rm C}) + nE(\mathrm{He}_1) - E(\mathrm{He}_n X) - (n_{\rm W} + n_{\rm C} + n - 1)E_{\rm bulk} \tag{4}$$

$$E_{\rm int}(d) = E(\text{He–He pair at } d) + E_{\rm bulk} - 2E(\mathrm{He}_1) \qquad (\text{positive} = \text{repulsive}) \tag{5}$$

For $n = 0$, $E_{\rm b}^{\rm tot}$ is the binding of the empty cluster against isolated vacancies (positive = vacancies attract). Every expression has the same number of each species on both sides, so the chemical potentials cancel.

Selection. For each (cluster, n): $E = \min\{E_{t,c} : n_{\rm far} = 0\}$ over trials t and cycles c (if none is intact, the overall minimum, flagged `ALL-MUTATED`).

$$\text{hits} = \#\{(t,c) : E_{t,c} - E < \text{TOL}\}, \qquad \text{spread} = \max_t \left(\min_c E_{t,c}\right) - E,$$

$$\text{MUTATED-STATE-LOWER-BY} = E - \min_{\rm all} E_{t,c} \ (\text{if} > \text{TOL}), \qquad \text{ESCAPED} : k_{\rm in} < n, \qquad \text{SPREAD} : R_g - R_{g0} > 1.5\,\text{Å}.$$

5.2 Diagnostics computed in the decks

$$d_i = |\mathbf{r}_i - \mathbf{r}_i^0| \text{ (host atom } i\text{)}, \qquad d_{\max} = \max_i d_i, \qquad n_{\text{far}} = \textstyle\sum_i [\, d_i > 1\,\text{Å}\,],$$

$$R_g = \left(\tfrac{1}{n}\textstyle\sum_j |\mathbf{r}_j - \mathbf{r}_{\text{cm}}|^2\right)^{1/2} \text{ (He)}, \qquad k_{\text{in}} = \#\{\text{He} \in \text{core}\}, \qquad r_{\text{core}} = \max_s |\mathbf{s} - \bar{\mathbf{s}}| + 2.2\,\text{Å}.$$

5.3 Geometry (`make_v3.py`)

- Minimum image: $\Delta x \leftarrow \Delta x - L_x\,\text{round}(\Delta x/L_x)$ (same for y, z); distances use it everywhere.
- Cluster growth: the new site of species s is the free lattice site minimising $\sum_{\mathbf{p}\in X'} |\mathbf{r} - \mathbf{p}|$ (compact; ties → lowest coordinates). For `vw2` this gives the c-axis neighbour (2.815 Å, since $c < a$); `vw2a` is restricted to the same basal plane (2.919 Å).
- He placement: uniform in spheres of radius R_{site} around the vacated sites, accepted if every He–He ≥ 1.5 Å and He–host ≥ 1.5 Å; on failure the limits relax stepwise (He–He to 1.0, He–host to 1.2 Å, radius +0.2, +0.4, +0.6 Å).
- Positions are written in lattice-cell units of the restart box, $x = x_{\text{lo}} + f_x L_x/N_x$ ($N_x = 8$), so they follow the relaxed cell exactly.

5.4 bcc-W benchmark (JW 2013 conventions)

$$E_{\text{f}}(V) = E(V) - \tfrac{N-1}{N}E_{\text{bulk}}, \qquad E_{\text{f}}(\text{SIA}) = E(\text{SIA}) - \tfrac{N+1}{N}E_{\text{bulk}} \tag{6}$$

$$E_{\text{f}}(\text{He}, X) = E(\text{He}X) - E(X) - \varepsilon_{\text{He}} \quad (\varepsilon_{\text{He}} = 0 \text{ isolated, or } -0.00538 \text{ eV fcc He}) \tag{7}$$

$$E_{\text{f}}(\text{He}_n V_m) = E - (N-m)\,\varepsilon_{\text{W}} - n\,\varepsilon_{\text{He}} \quad (\text{JW eq. 4: formation vs bulk W and fcc He}) \tag{8}$$

$$E_{\text{b}}(A \to B) = E_{\text{f}}(A) + E_{\text{f}}(B) - E_{\text{f}}(AB) \quad (\text{JW eq. 5; identical to the total-energy form when atoms balance}) \tag{9}$$

$$E_{\text{m}} \approx E(\text{He held at tet–tet midpoint}) - E(\text{He}_{\text{tet}}), \qquad E_{\text{dt}} = E_{\text{b}} + E_{\text{m}} \tag{10}$$

Vacancy-cluster rule (JW): relax V_m, compute per-atom energy, remove the W atom with the highest energy (ties → lowest ID), relax → V_{m+1}. Benchmark score: MAE $= \frac{1}{N}\sum |E^{\text{this}} - E^{\text{JW}}| = 0.015$ eV (N=46), max 0.072 eV.

5.5 Formation energies (`analyze_ef.py`)

$\Delta E(d) = E(d) - E_{\text{bulk}}$; d_{W}, d_{C} = atoms *added* (a vacancy has −1).

$$E_f(d) = \Delta E(d) - d_{\text{W}}\,\mu_{\text{W}} - d_{\text{C}}\,\mu_{\text{C}}, \qquad \mu_{\text{W}} + \mu_{\text{C}} = \varepsilon_{\text{WC}} = E_{\text{bulk}}/N_{\text{f.u.}} \tag{11}$$

$$\mu_{\text{C}} \le \varepsilon_{\text{C}}, \qquad \mu_{\text{W}} \le \min\big(\varepsilon_{\text{W}},\ \varepsilon_{\text{W}_2\text{C}} - \varepsilon_{\text{WC}}\big) \quad (\text{W2C term only if } \varepsilon_{\text{W}_2\text{C}} < \varepsilon_{\text{WC}} + \varepsilon_{\text{W}}) \tag{12}$$

$$\Delta H_f(\text{WC}) = \varepsilon_{\text{WC}} - \varepsilon_{\text{W}} - \varepsilon_{\text{C}} \quad (= \text{window width when bcc W sets the W-rich limit}) \tag{13}$$

$$E_S = \Delta E(V_{\text{W}}) + \Delta E(V_{\text{C}}) + \varepsilon_{\text{WC}}, \quad E_{FP}^{\text{C}} = \Delta E(V_{\text{C}}) + \Delta E(C_i), \quad E_{\text{b}}(X{+}Y) = E_f(X) + E_f(Y) - E_f(XY) \tag{14}$$

$$c_d = g_d\,\frac{e^{-E_f/k_BT}}{1 + e^{-E_f/k_BT}}, \qquad \frac{N_{\text{C}}}{N_{\text{W}}} = \frac{1 - c_{V_{\text{C}}} + c_{C_i} - c_{W_{\text{C}}} + c_{C_{\text{W}}} - 2c_{V_{\text{C}}^2} - c_{V_{\text{W}}V_{\text{C}}}}{1 - c_{V_{\text{W}}} + c_{W_i} + c_{W_{\text{C}}} - c_{C_{\text{W}}} - c_{V_{\text{W}}V_{\text{C}}}} \tag{15}$$

$$\text{50/50: } \frac{N_{\text{C}}}{N_{\text{W}}}(\mu_{\text{C}}, T) = 1, \qquad \text{WC}_{1-y}: \ \frac{N_{\text{C}}}{N_{\text{W}}}(\mu_{\text{C}}, T) = 1 - y \quad (\text{bisection; } N_{\text{C}}/N_{\text{W}} \text{ rises with } \mu_{\text{C}}) \tag{16}$$

$$E_f(\text{He}_n X) = E(\text{He}_n X) - E_{\text{bulk}} + n_{\text{W}}\mu_{\text{W}} + n_{\text{C}}\mu_{\text{C}} \quad (\mu_{\text{He}} = 0: \text{ He from vacuum}) \tag{17}$$

5.6 Dynamics used by the search

Berendsen: $\lambda = \left[1 + \frac{\Delta t}{\tau}\left(\frac{T_{\text{target}}}{T} - 1\right)\right]^{1/2}$, $\mathbf{v} \leftarrow \lambda \mathbf{v}$;

$T = \dfrac{2E_{\text{kin}}}{(3N-3)\,k_{\text{B}}}$;

quench: $T_{\text{target}}(t) = T + (1-T)\,t/t_{\text{run}}$;

CG stop: $\|\mathbf{F}\|_2 = \left(\sum_{i=1}^{N} |\mathbf{F}_i|^2\right)^{1/2} < \text{ftol}$.

6 Glossary

ABOP / Tersoff analytical bond-order potential: bond strength depends on the local environment. `tersoff/zbl` adds a screened-Coulomb (ZBL) wall at short range.

hybrid/overlay several pair styles summed for the same atom pairs; here host ABOP + He tables.

Pair table potential given as $E(r)$, $F(r)$ points; LAMMPS interpolates (`linear`, N points).

E_{bulk} energy of the perfect relaxed cell, same box as every defect cell.

Solution energy E_{sol} cost of putting one He from vacuum into the perfect lattice.

Formation energy energy of a defect relative to reference reservoirs (bulk W/WC, He). In WC needs $\mu_{\text{W}}, \mu_{\text{C}}$.

Binding energy E_{b} energy released when two defects combine; positive = bound.

Interaction energy E_{int} pair energy relative to two isolated He; positive = repulsive.

Trap defect that lowers He energy (vacancy, vacancy cluster).

Capacity largest n with $E_{\text{b}}(\text{He} \to \text{He}_{n-1}X) > 0$.

Trap mutation He pressure pushes a host atom off its site, making a Frenkel pair; detected by $n_{\text{far}} > 0$.

Channel / trigonal-prismatic site empty prism between W layers in WC (one per C), stacked along c; lowest-energy interstitial He site.

V_{W}, V_{C} W, C vacancy; $(V_{\text{W}})_2$ c-axis or in-plane divacancy; $V_{\text{W}}V_{\text{C}}$ mixed.

Tetrahedral / octahedral (bcc) interstitial sites in W; He prefers tetrahedral.

SIA self-interstitial atom (in W the ⟨111⟩ dumbbell/crowdion).

Constant-volume convention defect cells keep the bulk box; standard for binding energies (JW).

CG conjugate-gradient minimisation; **line search** picks the step along each direction.

ftol / etol force (2-norm, eV/Å) and relative energy stopping criteria.

Anneal / quench heat at T to cross barriers, then cool to ∼1 K before CG.

Berendsen thermostat rescales velocities toward a target T with time constant τ; good for searching, not for sampling.

Configuration search finding the lowest-energy arrangement of He (and host) among many local minima.

Trial one independent random He placement; **cycle** 1–4 = relaxation as placed / after 300 / 614.6 / 900 K.

hits relaxations reaching the reported minimum within 1 meV.

Intact no host atom moved > 1 Å.

k_{in}**, core** He inside the sphere (farthest site +2.2 Å); fewer than n = escaped.

$d_{\max}$, n_{far} largest host displacement; host atoms moved > 1 Å.

R_g He radius of gyration; jump vs start = He spread out.

Minimum image shortest vector between two points under periodic boundaries.

Restart file binary snapshot of box, atoms and settings (`wc_bulk0.restart`).

Compute currency (29 Sep 2021) computes are only valid on the timestep they were last evaluated.

`loop geom` velocities generated from positions, independent of IDs and MPI ranks.

Migration energy E_{m} barrier for a jump; **detrapping** $E_{\text{dt}} = E_{\text{b}} + E_{\text{m}}$.

NEB nudged elastic band: finds the minimum-energy path and barrier between two minima.

DFT column `YES` must, `rec` recommended, `check` compare with literature, `no` not needed.

MAE mean absolute error against the JW reference values.

JW 2013 Juslin & Wirth, J. Nucl. Mater. 432 (2013) 61 – the benchmark.

Chemical potential μ_s energy per atom of the reservoir that exchanges species s with the crystal; sets formation energies.

W-rich / C-rich limit WC in equilibrium with bcc W (or W2C) / with graphite: the two ends of the μ window.

Formation enthalpy ΔH_f energy of WC relative to bcc W + graphite; the width of the μ window.

Convex hull / coexistence the set of stable phases; at a two-phase boundary both phases fix μ.

Schottky pair one V_{W} + one V_{C}: removes a whole formula unit, so independent of μ.

Frenkel pair vacancy + interstitial of the same species.

Antisite an atom on the other species' site (W_{C}, C_{W}).

Dumbbell two atoms sharing one lattice site, split along a direction.

Site fraction c_d defects per formula unit at equilibrium (dilute: $\ll 1$).

50/50 (stoichiometric) μ the μ_{C} at which equilibrium defects keep C/W = 1; clamped to

the window if impossible.

Mid-window the midpoint of the μ_C window, a common convention.

WC_{1-y} C-deficient WC; beyond the W-rich limit it is a non-equilibrium (e.g. sputter-depleted) state.

Reference phase crystal that fixes a chemical potential, computed with the same potential.

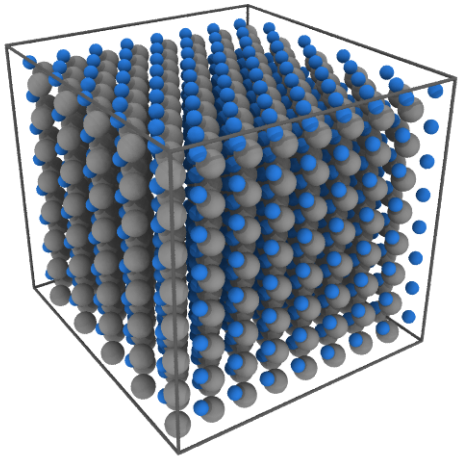
7 Renders of the simulation cells

Ray-traced with OVITO (`scripts/renders.py`); your own relaxed minima can be drawn the same way with `scripts/render_dump.py dumps/<file> --cfg cfg/<file>`. W gray, C blue, He yellow, vacancy red, added atom green.

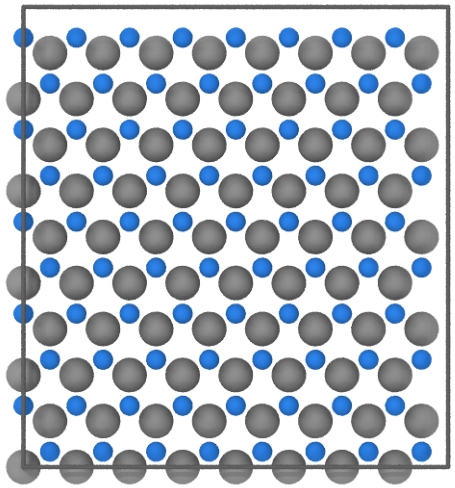
WC simulation cell (all v3 and formation-energy runs)

hexagonal WC in an orthogonal a x sqrt(3)a x c cell; periodic in x, y, z; 23.35 x 25.28 x 22.52 A at the 0 K lattice

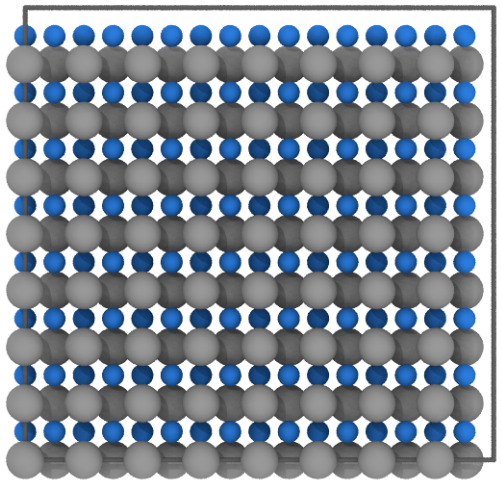
3D view
8 x 5 x 8 orthogonal cells, 1280 atoms (640 W + 640 C)



view down c [0001]
23.35 x 25.28 A; W-W in-plane a = 2.9186 A



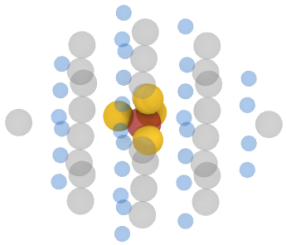
view down y
W layers (z = 0) and C layers (z = c/2), c = 2.8150 A



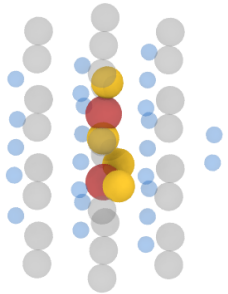
v3 vacancy clusters, as placed by make_v3.py

host atoms within 5.2 A of the cluster centre, small and translucent; red = vacated site, yellow = He (4 He, trial 1 of 4)

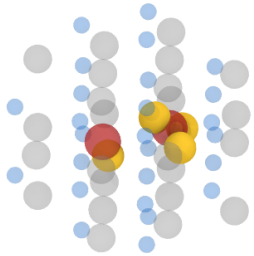
V_W + 4 He
vw1, trial 1, as placed (before CG)



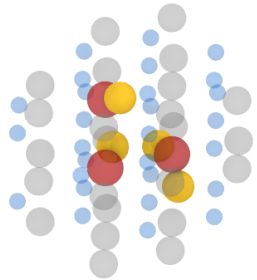
(V_W)2, c-axis + 4 He
vw2, trial 1, as placed (before CG)



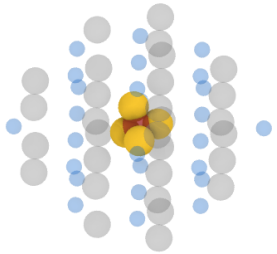
(V_W)2, in-plane + 4 He
vw2a, trial 1, as placed (before CG)



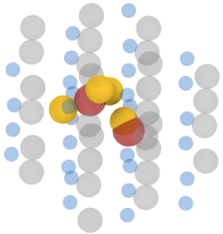
(V_W)3 + 4 He
vw3, trial 1, as placed (before CG)



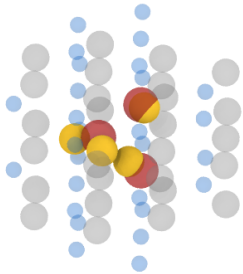
V_C + 4 He
vc1, trial 1, as placed (before CG)



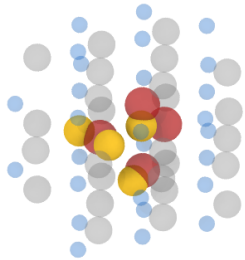
V_W V_C + 4 He
vwc, trial 1, as placed (before CG)



V_W (V_C)2 + 4 He
vwc2, trial 1, as placed (before CG)



(V_W)2 (V_C)2 + 4 He
vw2c2, trial 1, as placed (before CG)

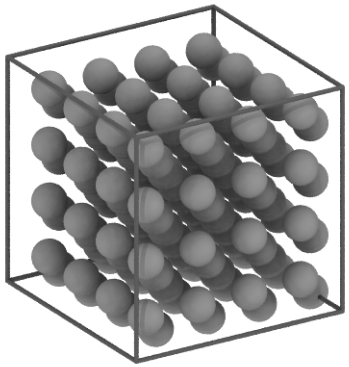


W C He vacancy added atom (interstitial / antisite)

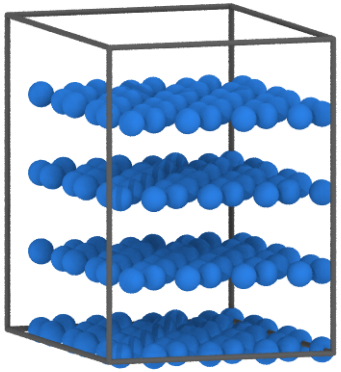
Reference crystals of in.phases (same potential as WC)

drawn at the box your run relaxed to; they fix the chemical-potential window for the formation energies

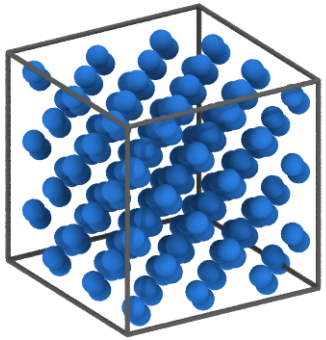
bcc W
a = 3.1650 A; eps_W = -8.8901 eV/atom; W-rich limit



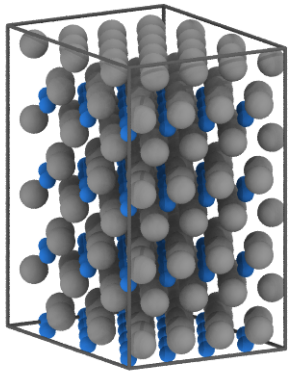
graphite (= graphene here)
a = 2.516 A; eps_C = -7.3612 eV/atom; C-rich limit



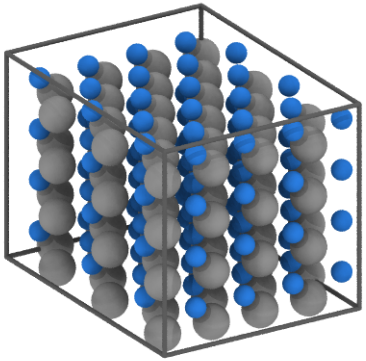
diamond
a = 3.5601 A; 45 meV/atom above graphite



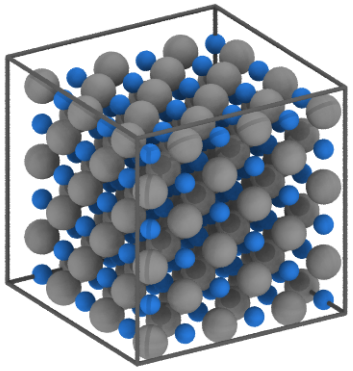
W2C (CdI2 type)
a = 2.985, c = 5.041 A; +1.59 eV/f.u. above WC + W



hexagonal WC
a = 2.9186, c = 2.8150 A; dH_f = -0.409 eV/f.u.



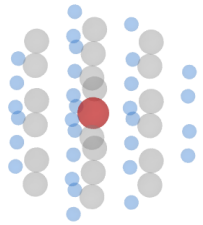
rocksalt WC
a = 4.383 A; +0.98 eV/f.u. above hexagonal



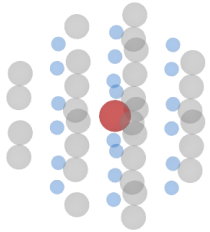
Formation-energy defect cells, as built by make_ef.py

host within 5 A, small and translucent; formation energies from your run (W-rich / C-rich, eV)

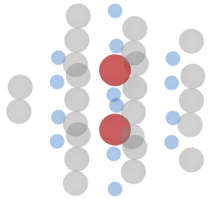
V_W
E_f 3.67 (W-rich) / 3.26 (C-rich)



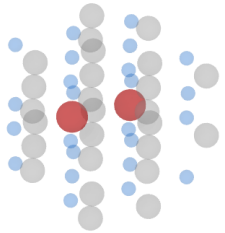
V_C
E_f 0.52 / 0.93



V_C pair, c-axis
binding -0.46 eV (repel)



V_C pair, in-plane
binding -0.25 eV (repel)



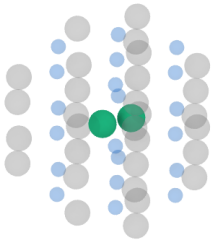
V_W + V_C
E_f 3.81 (any mu); binding +0.38



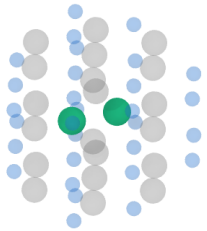
C_i, channel site (lowest)
E_f 3.22 / 2.81; same site as He



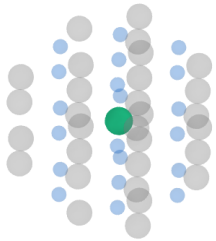
C_i, C-C dumbbell along a
0.79 eV above the channel site



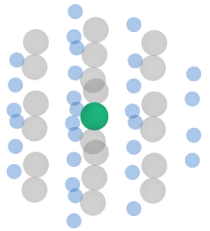
W_i, W-W dumbbell along a
E_f 12.34 / 12.74 (relaxes off-site)



W on a C site
E_f 9.22 / 10.03



C on a W site
relaxes toward the channel; 6.75 / 5.93



W C He vacancy added atom (interstitial / antisite)